

Divye Kalra

Livermore, CA | divyekalra3@gmail.com | +1 4438006625 | [LinkedIn](#) | [Github](#) | [Website](#)

EDUCATION

Johns Hopkins University

M.S. Cybersecurity

Birla Institute of Technology and Science (BITS) Pilani

B.E. Electrical and Electronics; M.Sc. Mathematics (Double Major)

Baltimore, USA

Aug 2024 – Dec 2025

Hyderabad, India

Aug 2019 – Jun 2024

TECHNICAL SKILLS

- **Languages & Core Technologies:** Python, C, C++, Java, Rust, SQL, Bash, HTML/CSS, AWS, Azure, GCP, Docker, Kubernetes, GKE, OpenStack, Linux, FastAPI, Django, Burp Suite, Wireshark, Ghidra, Git, GitHub, MySQL, PostgreSQL, Prometheus, Grafana, Loki, Tempo, Proxmox, Jira, Ollama, Terraform, GitOps
- **Security:** Application Security, Secure SDLC, Threat Modeling, SIEM, Incident Response, Logging and Monitoring, Threat Intelligence, IAM/RBAC/MFA/SSO, Endpoint Security, Zero Trust Architecture, Risk Management and Compliance, MITRE ATT&CK, ISO 27001, CIS, HIPAA, PCI-DSS

EXPERIENCE

GlobalLogic - Software Engineer Intern

June 2026 – Present

- Architected a GitOps-driven CI/CD platform on GKE using Terraform and Flux — eliminating an 11-step manual deployment process and delivering full-stack observability (Prometheus, Grafana, Loki, Tempo) across MinIO, Neo4j, Airflow, and Elasticsearch, with automated TLS via ZeroSSL and DNS configuration.

Wildlife for All — Information Security Engineer

Feb 2026 – June 2026

Bringing formal security oversight to a nonprofit's SaaS stack — auditing platforms staff rely on daily against CIS benchmarks and closing the highest-risk gaps first.

- Audited Google Workspace and Slack against CIS benchmarks, documenting 40+ misconfigurations across MFA, SPF/DKIM/DMARC, OAuth integrations, external sharing, and offboarding — each mapped to a CIS control with severity and remediation steps.
- Remediated highest-severity findings first: enforced admin MFA, removed unused OAuth connectors from 65 Slack integrations, restricted Drive public sharing, and progressed Gmail DMARC toward enforcement — closing the top account-takeover and data-exfiltration paths.

JHU Applied Physics Laboratory — Information Security Engineer

Jan 2025 – Dec 2025

- Designed and implemented an SCMS (Secure Credential Management System) reference in python — a 6-service system with a 3-tier CA hierarchy, signed V2X message broadcast, and a roadside server running 15+ real-time misbehavior checks against the SAE J3161 standard.
- Mitigated replay and location-spoofing attacks in V2X networks via a pseudonym certificate revocation pipeline using cryptographic linkage values; integrated C-V2X into an open-source platform, cutting CRL delivery latency by 20% vs. baseline DSRC broadcast.

IITB Trust Lab — Applied Cryptography Engineer

Jan 2024 – Jun 2024

- Delivered the first open-source Rust implementation of [CASE](#), cutting encryption overhead 30% via a hybrid symmetric/asymmetric fixed-size key design and accelerating anonymous signature verification 15% by batching hash computations and eliminating redundant scalar multiplications.

Cyber Security Hub, Macquarie University — Confidential Computing Engineer

Jul 2023 – Dec 2023

- Built a privacy-preserving record matching system using Trusted Execution Environments (TEEs) and differential privacy (DP) - achieving **0.45s** latency on 30K records and **~16s** on 1M records.
- Reduced side-channel information leakage by **35–45%** through distributed enclave partitioning, validated on datasets (size: 30K – 1M records); produced the first algorithm combining TEEs, multi-party computation, and DP for record matching

PROJECTS

– Production-Style Homelab Infrastructure

Fully self-hosted home infrastructure — deploying **10+** services across Proxmox VMs and containers including TrueNAS, Pi-hole, OPNsense, and Prometheus/Grafana/Uptime Kuma monitoring — with Tailscale VPN for remote LAN access without port-forwarding and select workloads on AWS, eliminating all cloud storage subscriptions.

– LLM-Powered Security Code Review Assistant

Built and shipped a two-phase AI agent pipeline — regex pattern matching to surface candidates, GPT to reason about exploitability in context and generate secure fixes — detecting 25+ vulnerability classes across 8+ languages at 95% accuracy; exposed as a customer-facing FastAPI web UI and CLI with every finding mapped to OWASP, CWE, MITRE ATT&CK, and CVSS.

– SBOM-Driven Runtime Admission Controller

Engineered a supply chain security pipeline spanning build-time SBOM generation, a Python admission webhook that intercepts pod creation and cross-references image SBOMs against EPSS exploitability scores, and dynamic OPA/Kyverno policy generation with threat modeling against XZ Utils-style supply chain attacks. Integrated Sigstore/Cosign for SBOM signing and verification.

– Kubernetes Security (Policies, Runtime, and Exploitation)

Executed full Kubernetes attack chains — Server-Side Request Forgery (SSRF) to steal a mounted service account token and enumerate cluster secrets, misconfigured RBAC to exfiltrate secrets via pod volume mount, and a privileged container escape to gain access to the underlying nodes — then hardened each attack path using Kyverno (policy-as-code) admission policies aligned to CIS benchmarks and validated detection coverage with Cilium Tetragon kernel-level eBPF telemetry.

CERTIFICATIONS AND ACHIEVEMENTS

- AWS Certified Cloud Practitioner (AWS CCP)
- CompTIA Security+